Multimedia Signal processing

Prof. Marco Tagliasacchi

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1. **Question (8 pts)**:
   Consider a continuous source X, whose p.d.f. is given by

$$p_X(x) = \frac{1}{2} e^{-|x|} \tag{1}$$

   (a) Design the Lloyd-Max quantizer for the source, using $M = 4$ quantization bins. Initialize the algorithm using the following thresholds $\{-\infty, -1, 0, 1, +\infty\}$. Stop after the first *complete* iteration.

   (b) Compute the entropy, in bits / symbol, of the source representing the output of the Lloyd-Max quantizer.

   Hint. You might find the following indefinite integral useful

$$\int a e^{ca} da = \frac{e^{ca}}{c^2}(ca - 1) \tag{2}$$

2. **Question (5 pts)**
   (a) Illustrate the design of analysis-by-synthesis speech coders.

   (b) Discuss the use of long- and short-term predictors.

3. **Question (5 pts)**
   (a) Illustrate the rate-distortion curve for discrete and continuous sources, specifying the achievable region. Express the distortion in terms of quantization SNR.

   (b) Compare different scalar quantization strategies in terms of their operational rate-distortion curves.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**
   (a) Consider a discrete source
   - [T] [F] $H_N(x) \geq H_{N+1}(x)$, where $H_N(x) = \frac{H(x_1,\ldots,x_N)}{N}$
   - [T] [F] $H_\infty(x) = H(x)$ iff the source is memoryless.
   - [T] [F] The average code length of an Huffman code L_H is such that $H(x) \leq L_H < H(x) + 1$
   - [T] [F] In a Huffman code, the two longest codewords have always the same length.

   (b) Consider the KLT.
   - [T] [F] The basis functions are orthonormal.
   - [T] [F] The autocorrelation matrix of the transform coefficients is diagonal.
   - [T] [F] Given the KLT learned on an image collection, the coding gain of the KLT is higher than that of the DCT, for each image in the collection.

   (c) Consider an image coding system
   - [T] [F] Spatial redundancy is mostly exploited by means of DPCM.
   - [T] [F] Threshold quantization leads to higher coding efficiency than zonal quantization.
   - [T] [F] In JPEG, the chroma components are subsampled by a factor of 2 both horizontally and vertically.

5. **Question (Matlab - 10pts)**
   Consider the image *image.png*. Arrange the pixel values of the image in two vectors, $\mathbf{x}_r$ and $\mathbf{x}_c$, obtained by reading the image in raster scan order either row-wise or column-wise.
   (a) Design a DPCM coder to lossy code both $\mathbf{x}_r$ and $\mathbf{x}_c$. Use a scalar uniform quantizer, whose step size Δ can be adjusted, ignoring the saturation region.
   (b) By varying Δ, compute the operational rate-distortion curves for both $\mathbf{x}_r$ or $\mathbf{x}_c$. Estimate the rate as the entropy of source representing the output symbols of the quantizer.
   (c) Plot the rate-distortion curves (PSNR vs. rate) and comment the result.